QC STORY PROJECT

PROJECT TITLE :-

To Improve Productivity At Under Body 30 – Framing Line

Mahindra and Mahindra Ltd, Nashik Plant, Scorpio Body Shop

	Plan	Actual
Start Date	**01.08.12**	**09.08.12**
Completion Date	**10.12.12**	**30.11.12**

Current Status: **Closed**

TOKEN NO 209301
Rajesh Dhanokar

Cross Functional Team

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Step1. Select Subject

Brief description of Problem :

25% productivity Rise of Under Body #30 – Framing Line

For Ex : 91 BIW/Shift to 114 BIW/Shift (25%)

Step1. Select Subject

What : 25 % Productivity Rise

Where : Scorpio – Body Shop, Framing Line

When : November ‘12

Why : Promise 2013 (We will produce 2L Vehicles in a year)

Who : Productivity CFT

How : Thru de-bottlenecking of process (Work content Reduction)

Step 2. Justify Choice

1

Mahindra Nashik Plant

PROMISE 2013

- Achieve double digit 3 MIS for build quality of Nashik products. We shall ensure Zero PDI defects at dealer @ Zero KM
- We will produce 200,000 vehicles in a year.
- We will support marketing to meet the customer demand by delivering 100% vehicles as per e-DCM committed dates.
- We will be ready to produce 50,000 export vehicles.
- We will ensure ramp up of new products as per launch plan.
- We will strive for reducing consumption of natural resources used in our manufacturing process.
- Embracing innovative ways and new technologies to continually reduce the operational cost by 25% over F10.

We shall strive to achieve Level 6 in MQW

25% Productivity rise

Step 2. Justify Choice

2. Capacity enhancement.

3. Reduction in Non Value added Activity

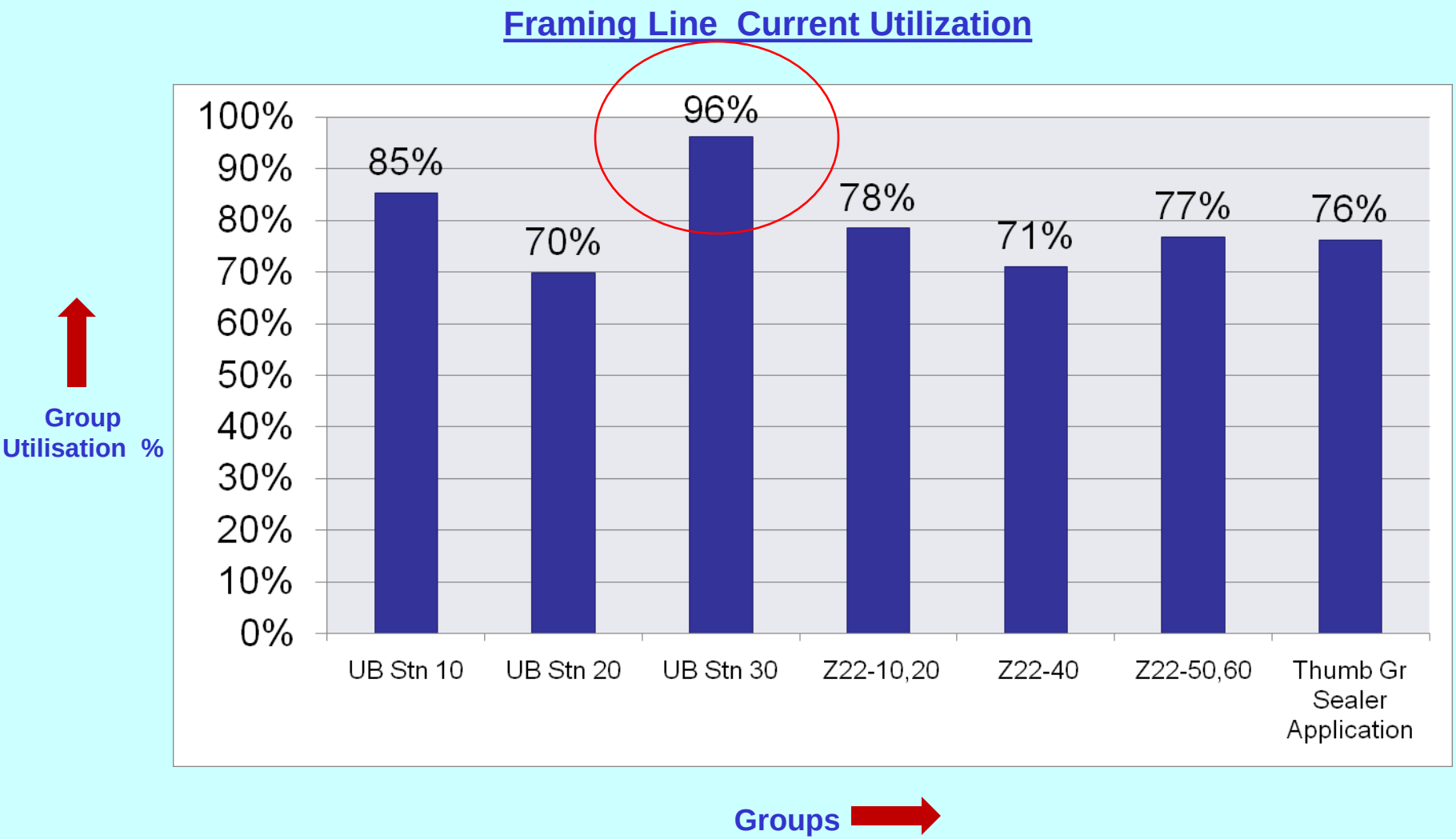
4. Long Term Agreement (LTA) preparation.

5. Increase in Equivalent vehicle per person per year

Step 2. a Immediate Containment Action

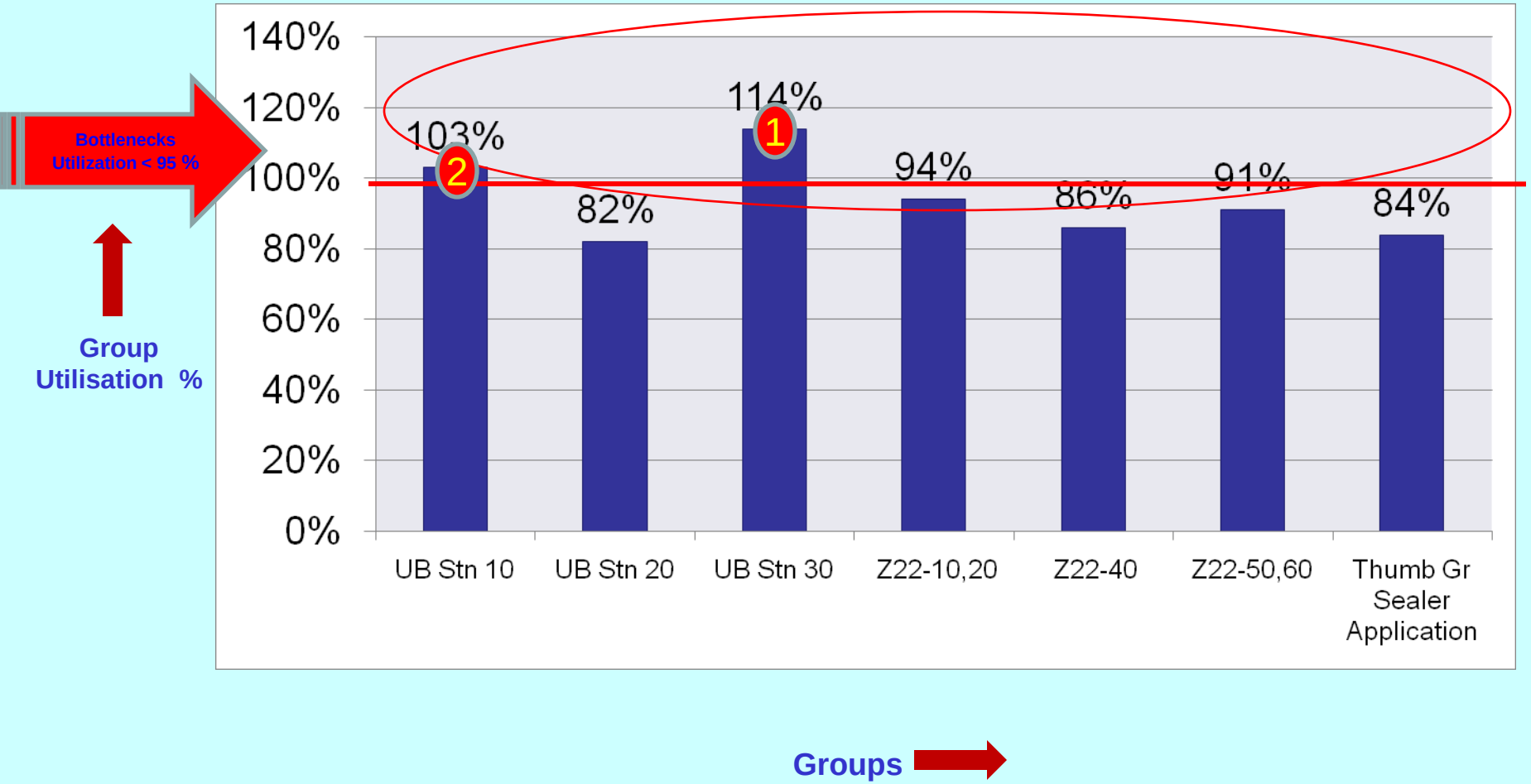
Not Applicable

Step 3. Understanding Current Situation



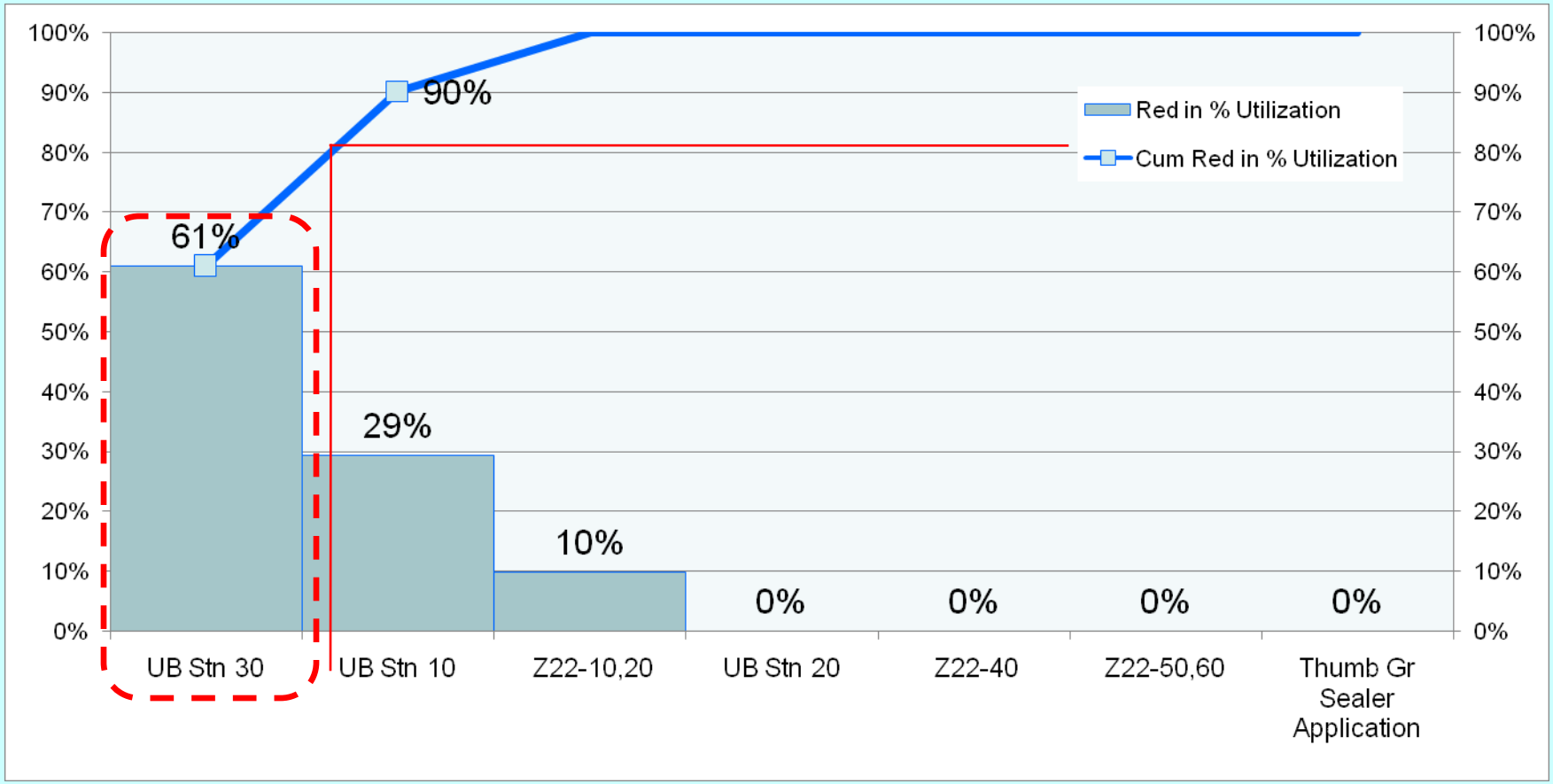
Step 3. Understanding Current Situation

Framing Line Utilization After 25% Rise@4.4miles/hr



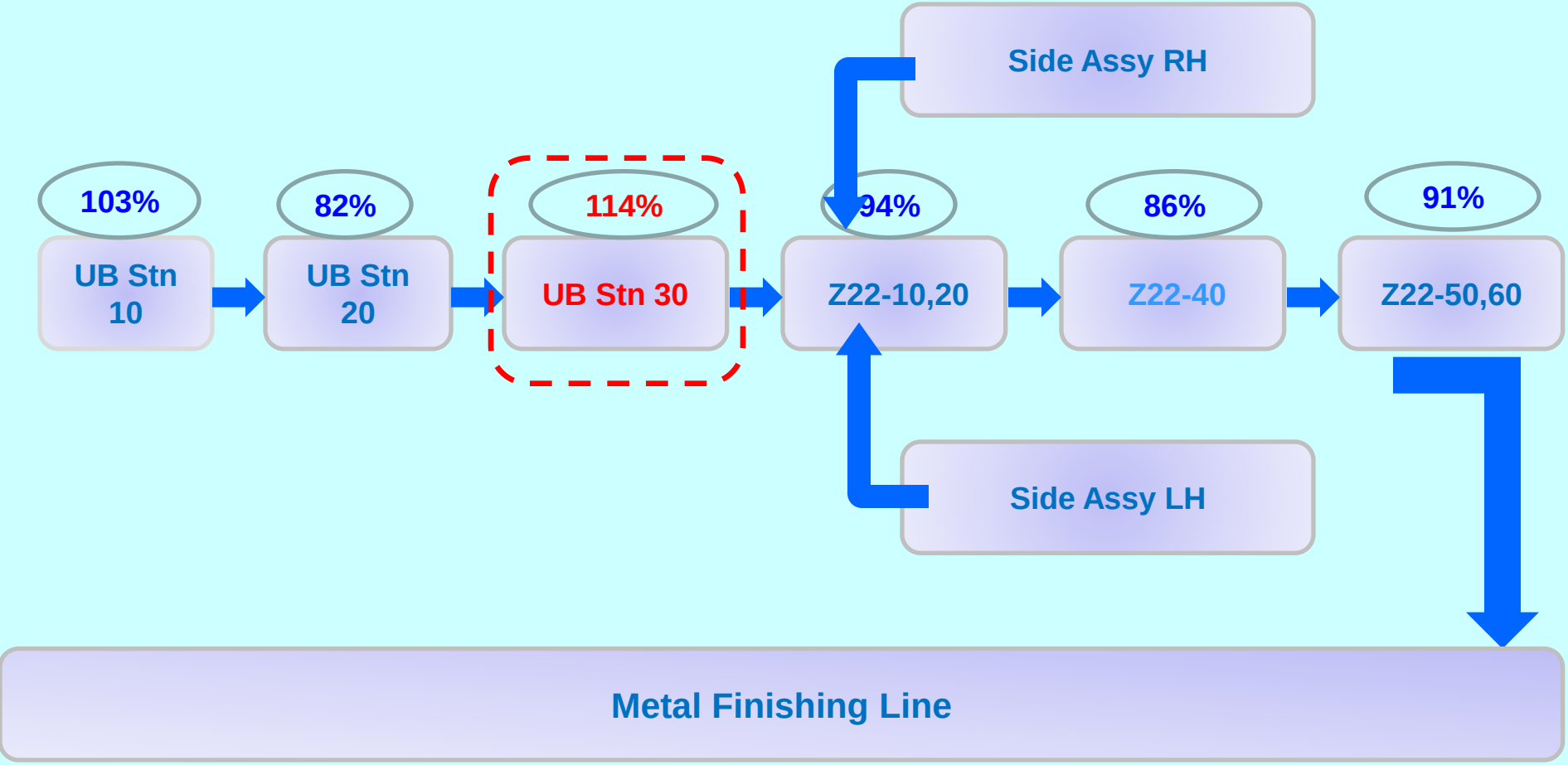
Step 3. Understanding Current Situation

Pareto Chart – Reduction in % Utilization Required



Step 3. Understanding Current Situation

Framing line - Process Flow Diagram with 25% Rise



Step 3. Understanding Current Situation

Work content study - Framing Line- Underbody Stn30

SN	Operation Description	Workcontent In Manminutes 3.8miles/Hr	Manpower Required	Manpower Deployment	Utilisation %
	Underbody Station#30				
1	SPOT WELD (ASSEMBLY UNDER BODY & FRONT END) LH/RH OPERATOR [GUN X01]	2.5			
2	SEALER APPLICATION (ASSEMBLY UNDER BODY & FRONT END) RH & LH OPERATOR	0.5			
3	SEALER APPLICATION (AASY UBODY & FRONT END) W408 SEALER APPLICATION AT FRAMING LINE UB 30	0.2			
4	PLACE PARTS ON FIXTURE & PULL BACK X01 GUN [RH/LH]	0.6			
5	PLACE REAR FLOOR ON BODY	1.1			
6	LOADING 2 PARTS (EXTN QTR INR & REINF ANCHRG)+ SPOTWELDING AFTER LOADING RR FLOOR [C-04] [RH/LH]	2.3			
7	SPOTWELDING AFTER LOADING RR FLOOR [C-02] [RH/LH]	1.6			
8	SPOTWELDING ON REING ANCHRG PLR B [C-02] [RH/LH]	0.7			
9	SPOTWELDING AFTER LOADING RR FLOOR [X-24]+SPOTWELD BY PINCH GUN [RH/LH]	1.6			
10	GUN TIP FILING AT RR FLOOR SPOT WELDING ON FRAMING LINE [OP.NO.30] [RH/LH]	0.3			
11	TRAVEL CARD FILLING AT U/B STAGE NO.30	0.2			
12	SPOT WELDING BY X24 GUN (U/B 40) RH/LH	1.5			
13	GUN TIP FILING (U/B 40) (RH/LH)	0.1			
	Total	13.14			
	Setup - 114	@3.8M/Hr	3.75	3	125%
	Setup - 114	@4.4M/Hr	3.41	3	114%

Step 3. Understanding Current Situation

Operator wise Balancing

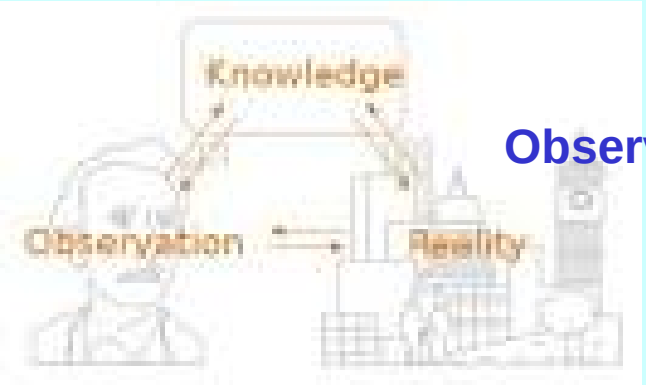
Group	Workcontent In Manminutes 3.8miles/Hr	Setup	Number Of Operators	1	2	3	4	5	Average Utilisation	Min Utilisation	Max Utilisation
Underbody Station#30	13.14	114	3	115	148	113	NA	NA	125	113	148

Step 3. Understanding Current Situation

Video study of the group operator – Framing Line Underbody Stn30

Group	Setup	TACT	Number Of Operators	Number Of Observations	Opr 1	Opr 2	Opr 3
Underbody Station#30	92	5.05	3	1	3.9	4.8	3.4
				2	3.8	4.9	4.0
				3	3.9	4.8	3.7
				4	4.0	4.9	3.7
				5	4.1	4.9	3.8
				Avg Video Time	3.9	4.9	3.7
				Worst	4.1	4.9	4.0
				% Rise Possible	19%	2%	21%
				Best	3.8	4.8	3.4
				% Rise Possible	25%	5%	33%
				As per WC	4.2	5.0	3.9
				Difference	0.4	0.2	0.6

Step 3. Understanding Current Situation



Observations from above study

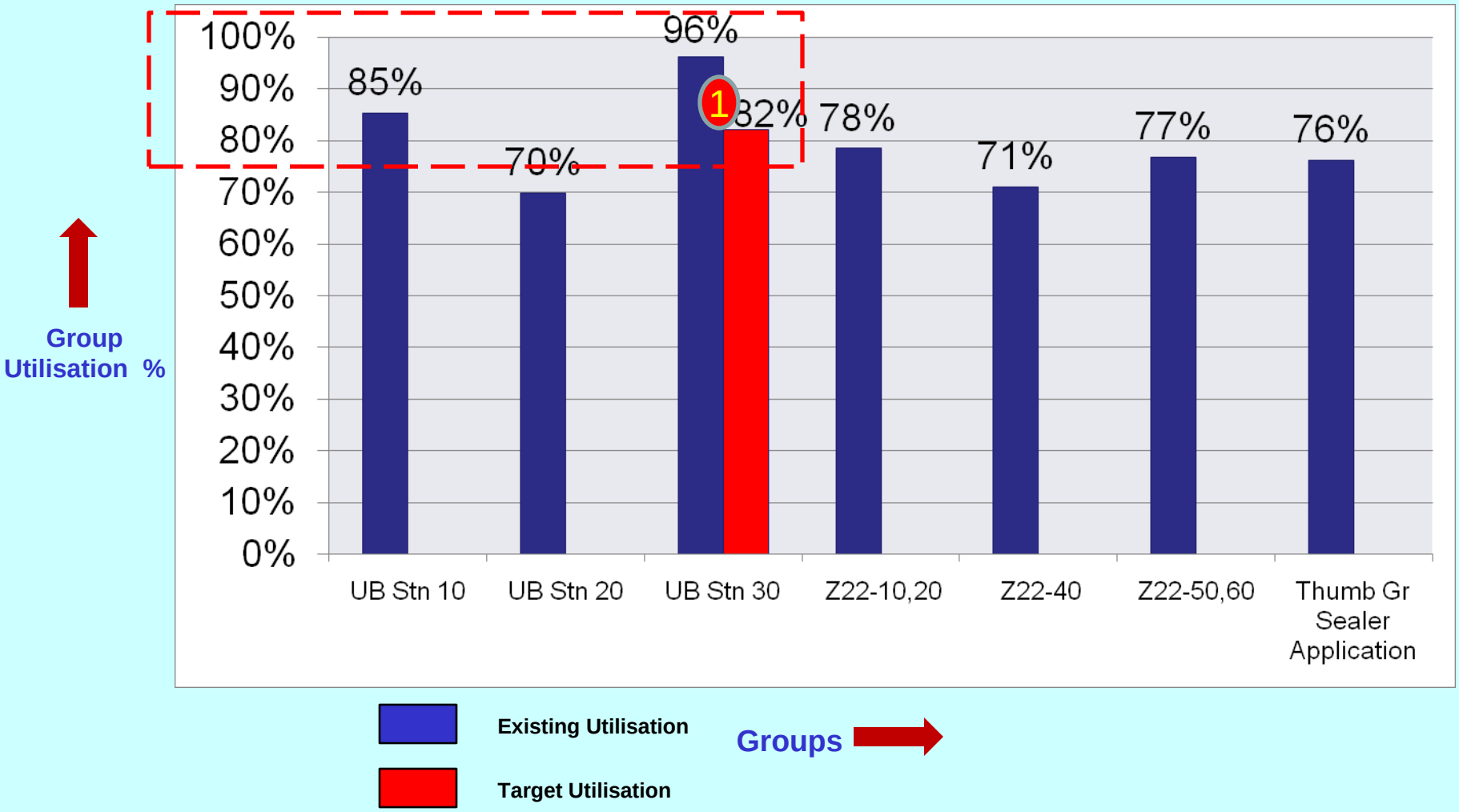
1. Difference in Most Work content and Video timings.

2. Scope for elimination of NVA like Removal of Protective cap, Plastic bags, Over Torqueing activities.

3. Scope for internal group Line balancing..

Step 4. Select Target

Framing Line Target Utilization



Step 5. Analysis

5.1 : Brainstorming –List of Possible causes

SNO	Possible causes
1	Wrong tool handling
2	Short cuts
3	Material collection Method
4	Untrained operators
5	Material storage
6	Bulky, Heavy parts
7	Work content not updated
8	Line balancing
9	SOS not updated
10	Over Processing
11	Layout/Facility modification
12	Machine time
13	Up gradation in facility

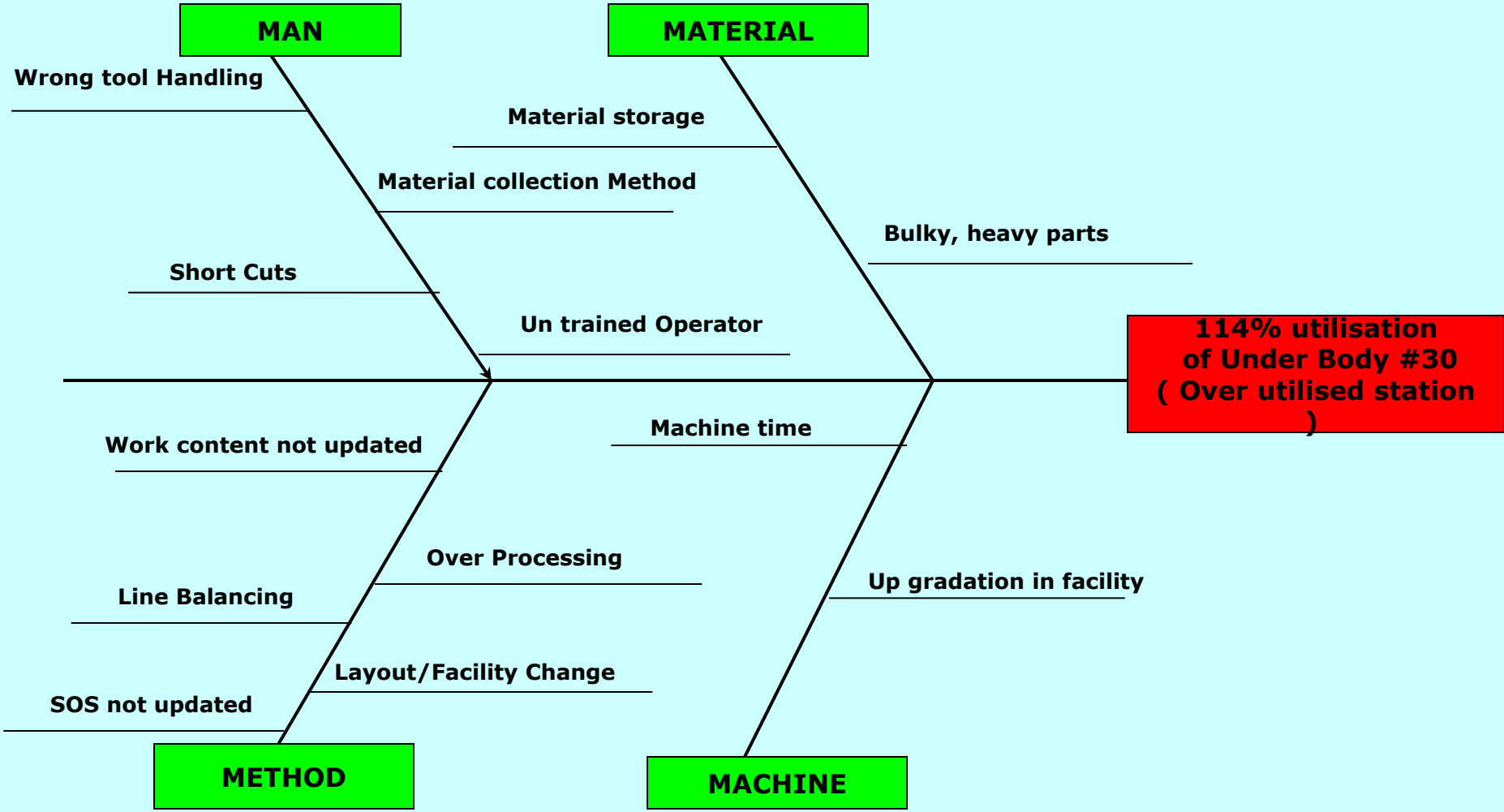
Step 5. Analysis

List of causes not considered as probable causes

SNO	Not considered Causes	Reasons
1	Untrained operator	Trained & Regular Operator
2	Up gradation in Facility	Capacity of 114 UPS
3	Bulky or heavy parts	As per Ergonomics study
4	SOS Not updated	SOS is dynamic documents
5	Layout/Facility Change	Capacity of 114 UPS

Step 5. Analysis

Possible Causes



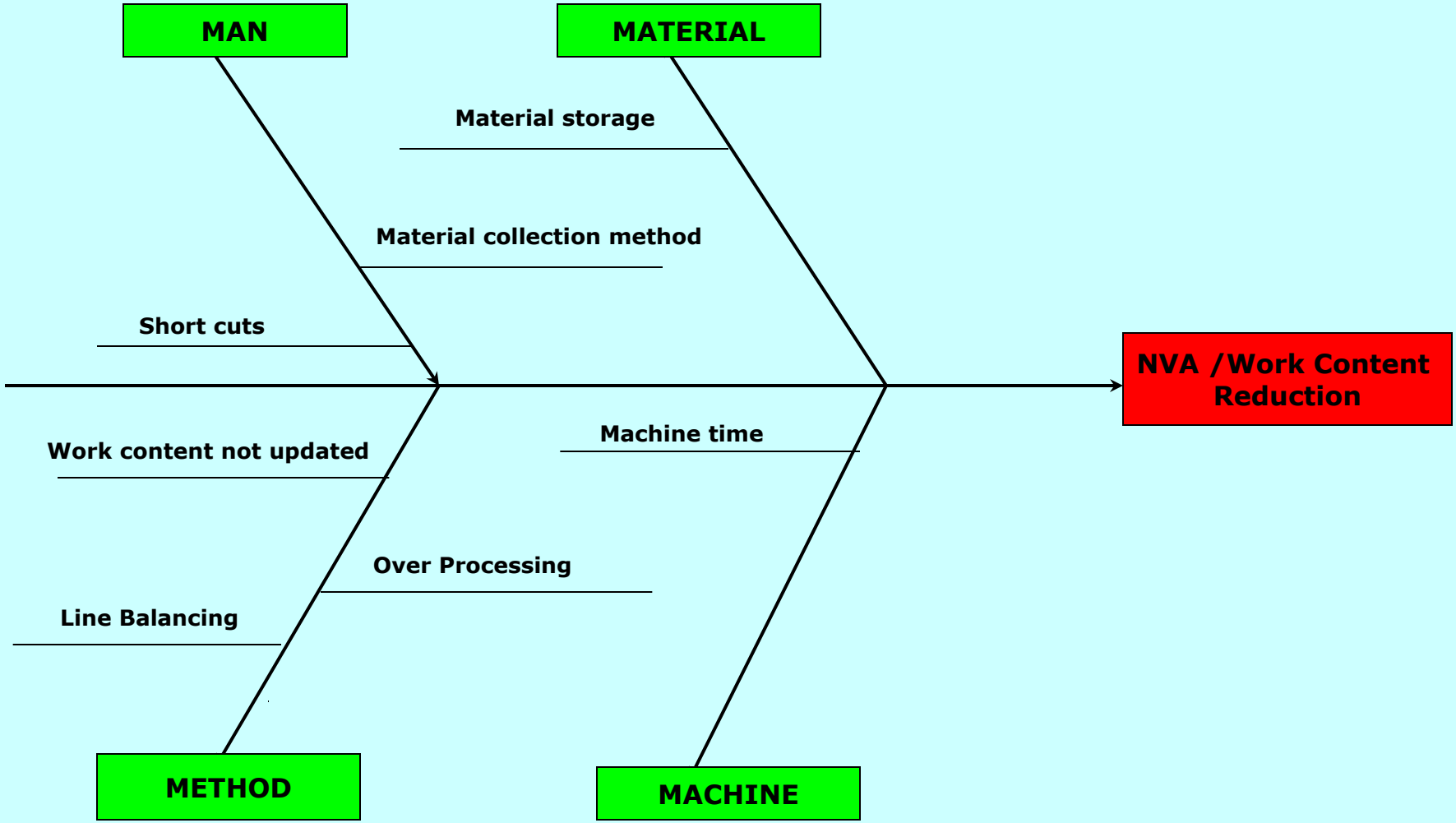
Step 5. Analysis

List of Probable Causes

SNO	Probable causes
2	Short cuts
3	Material collection method
5	Material storage
7	Work content not updated
8	Line balancing
10	Over Processing
12	Machine time

Step 5. Analysis

Probable Causes



Step 5.2 Test of Hypothesis

Confirm the probable cause:

S NO	Probable Cause	Testing and Observations	Conclusion
2 & 3	**2. Short Cut** **3. Material collection Method**	On Gemba it is observed that operators are using shortcuts to complete the task	Valid

For Ex :

SOS Steps :
1. Move 1-2 steps collect the **A** part
2. Move 1-2 steps back towards the body and put.
3. Move 1-2 steps collect **B** Part
4. Move 1-2 steps back towards the body and Put.

Actual Steps :
1. Move 1-2 steps collect A&B Parts
2. Move 1-2 steps back towards the body and put.

Step 5.2 Test of Hypothesis

Confirm the probable cause:

S NO	Probable Cause	Testing and Observations	Conclusion
5	**Material Storage**	Material collection and returns steps are more	valid

Operation Group MOST Elements

Group:

PART LOADING

OPN	SO. NO.	EL NO.	OPERATION / SUB OPERATION / ELEMENT DESCRIPTION	FREQ.	DIV. FREQ	MEN	OFF LINE	SUB OPERATION REFERANCE
5490			Tube to front LH Wheel to RH Wheel					
	1		MOVE 5-7 STEPS TO GRASP THE ANCHORAGE B PILLAR FROM THE CRATE	1	1	1	1	6-PHAND02
		1	MOVE 5-7 STEPS	1	1	1	No	
	2		GRASP PART FROM THE CRATE	1	1	1	1	6-PHAND02
		1	GRASP	1	1	1	No	
	3		MOVE 5-7 STEPS TWRDS BODY AND PUT ON FIXTURE	1	1	1	1	PHAND05
		1	MOVE 5-7 STEPS, PUT PART	1	1	1	No	
	4		BEND & ARISE FOR FITMENT	1	1	1	1	VBMOVE
		1	BEND & ARISE	1	1	1	No	
	5		MOVE 5-7 STEPS AND RR EXT OF REAR FLOOR	1	1	1	1	PHAND05
		1	MOVE 5-7 STEPS, PUT PART	1	1	1	No	

Step 5.2 Test of Hypothesis

Confirm the probable cause:

S NO	Probable Cause	Testing and Observations	Conclusion
7	**Work content /Element sheet not updated**	Revisited all the elements and Work content & found difference in Actual V/S Given	valid

SN	Operation Description	Workcontent In Manminutes 3.8miles/Hr
	Underbody Station#30	
1	SPOT WELD (ASSEMBLY UNDER BODY & FRONT END) LH/RH OPERATOR [GUN X01]	2.5
2	SEALER APPLICATION (ASSEMBLY UNDER BODY & FRONT END) RH & LH OPERATOR	0.5
3	SEALER APPLICATION (AASY UBODY & FRONT END) W408 SEALER APPLICATION AT FRAMING LINE UB 30	0.2
4	PLACE PARTS ON FIXTURE & PULL BACK X01 GUN [RH/LH]	0.6
5	PLACE REAR FLOOR ON BODY	1.1
6	LOADING 2 PARTS (EXTN QTR INR & REINF ANCHRG)+ SPOTWELDING AFTER LOADING RR FLOOR [C-04] [RH/LH]	2.3
7	SPOTWELDING AFTER LOADING RR FLOOR [C-02] [RH/LH]	1.6
8	SPOTWELDING ON REING ANCHRG PLR B [C-02] [RH/LH]	0.7
9	SPOTWELDING AFTER LOADING RR FLOOR [X-24]+SPOTWELD BY PINCH GUN [RH/LH]	1.6
10	GUN TIP FILING AT RR FLOOR SPOT WELDING ON FRAMING LINE [OP.NO.30] [RH/LH]	0.3
11	TRAVEL CARD FILLING AT U/B STAGE NO.30	0.2
12	SPOT WELDING BY X24 GUN (U/B 40) RH/LH	1.5
13	GUN TIP FILING (U/B 40) (RH/LH)	0.1

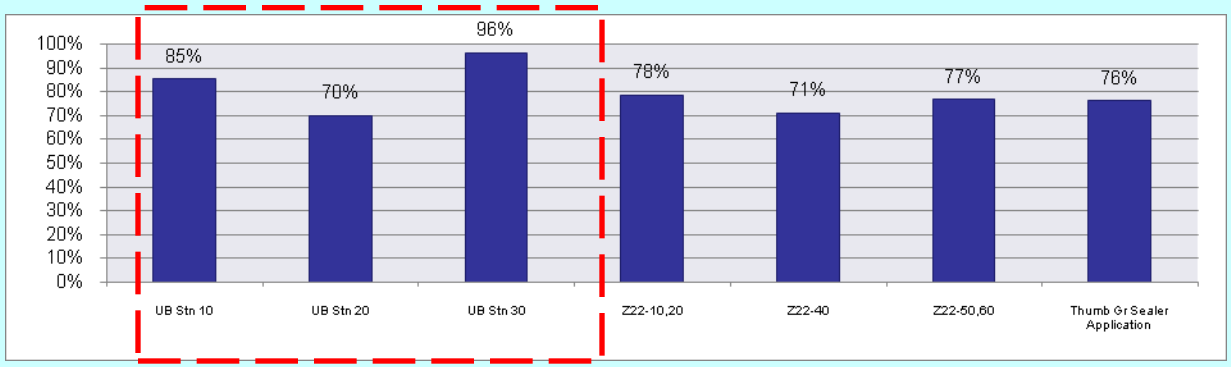
WC Changed

Step 5.2 Test of Hypothesis

Confirm the probable cause:

S NO	Probable Cause	Testing and Observations	Conclusion
8	**Line Balancing**	Group operator wise balancing and found that some operators are more utilized and some are less utilized	Valid

Group	Workcontent In Manminutes 3.8miles/Hr	Setup	Number Of Operators	1	2	3	4	5	Average Utilisation	Min Utilisation	Max Utilisation
Underbody Station#30	13.14	114	3	115	148	113	NA	NA	125	113	148



More utilized

Less utilized

Step 5.2 Test of Hypothesis

Confirm the probable cause:

S NO	Probable Cause	Testing and Observations	Conclusion
10	**Over Processing**	While revisiting the Elements found that many over processing activities are considered	Valid

Operation Group MOST Elements

Group:

PART LOADING

OPN	SO. NO.	EL NO.	OPERATION / SUB OPERATION / ELEMENT DESCRIPTION	FREQ.	DIV. FREQ	MEN	OFF LINE	SUB OPERATION REFERANCE
5490			**Tube to front LH Wheel to RH Wheel**					
	1		**MOVE 5-7 STEPS TO GRASP THE ANCHORAGE B PILLAR FROM THE CRATE**	**1**	**1**	**1**	**1**	**6-PHAND02**
		1	MOVE 5-7 STEPS	1	1	1	No	
	2		**GRASP PART FROM THE CRATE**	**1**	**1**	**1**	**1**	**6-PHAND02**
		1	GRASP	1	1	1	No	
	3		**MOVE 5-7 STEPS TWRDS BODY AND PUT ON FIXTURE**	**1**	**1**	**1**	**1**	**PHAND05**
		1	MOVE 5-7 STEPS, PUT PART					
	4		**BEND & ARISE FOR FITMENT**	**1**	**1**	**1**	**1**	**VBMOVE**
		1	BEND & ARISE	1	1	1	No	
	5		**MOVE 5-7 STEPS AND RR EXT OF REAR FLOOR**	**1**	**1**	**1**	**1**	**PHAND05**
		1	MOVE 5-7 STEPS, PUT PART	1	1	1	No	

Activity to be Challenged

Step 5.2 Test of Hypothesis

Confirm the probable cause:

S NO	Probable Cause	Testing and Observations	Conclusion
12	**Machine time**	Re visiting of Actual Process time and Given process time	In Valid

SNO	Critical Process	Actual Process time in Sec					Given PT Sec	Remarks
1	Spot welding Process time	1.08	1.08	1.08	1.08	1.08	1.08	
2	Retract Open / Close	1.08	1.08	1.08	1.08	1.08	1.08	
3	Rear Floor Hoist Up	12	12	12	12	12	12	
4	Rear Floor Hoist Down	12	12	12	12	12	12	
5	Shuttle Transfer time	30	30	30	30	30	30	

Machine time actual V/S Given Matching

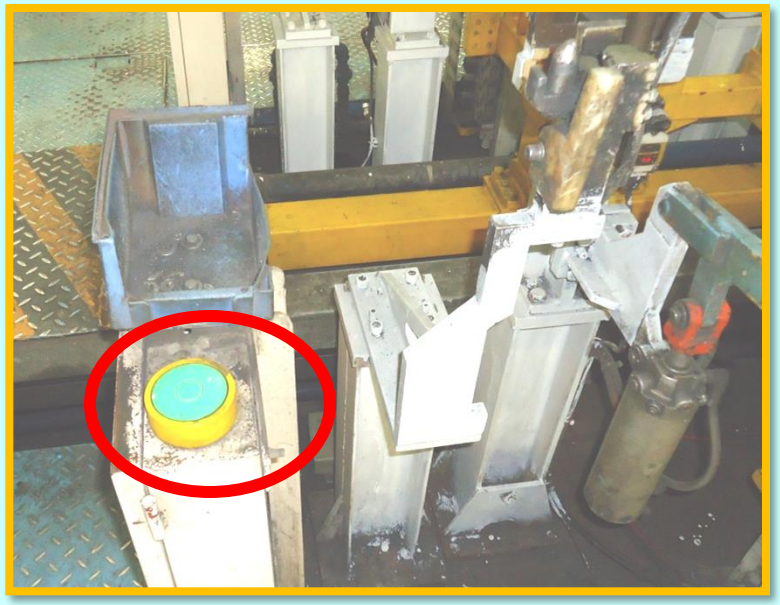
Step 6. Implement Action

Action Plan & Status @4.4miles/hr

SNO	Group	NVA /Work Content reduction in Min			Proposed Ideas for Reduction in Work content	Target	Responsibility	Status	Remarks
		WCR Required	WCR Identified	WCR Implemented					
1	Framing Line Under Body #30	0.5	0.8	0.5	Sealer application eliminated 2 points Eliminate 2 times open /close +push pull + turn gun Foot Padel Install – 4 nos (NVA reduction - B/A to be eliminated) NVA reduction - Put ext qtr inner & Anchor B plr in Batch of 10 on stage	10.12.12	Gokul/ Rajesh/ Sanket	Closed	After WCR : Current utilization : 82 % Future Utilization : **96 %** **(With Balancing)**

Step 6. Implement Action

Before



After

Before Improvement :

Bend & arise for operating fixture clamps

After Improvement :

Foot padel installed to eliminate bending

Step 6. Implement Action

Before

Before Improvement :

Sealer Application

After

After Improvement :

Sealer Eliminated

Step 6. Implement Action

Before

After

Before Improvement :

Transformer & guns on MS rails which required effort

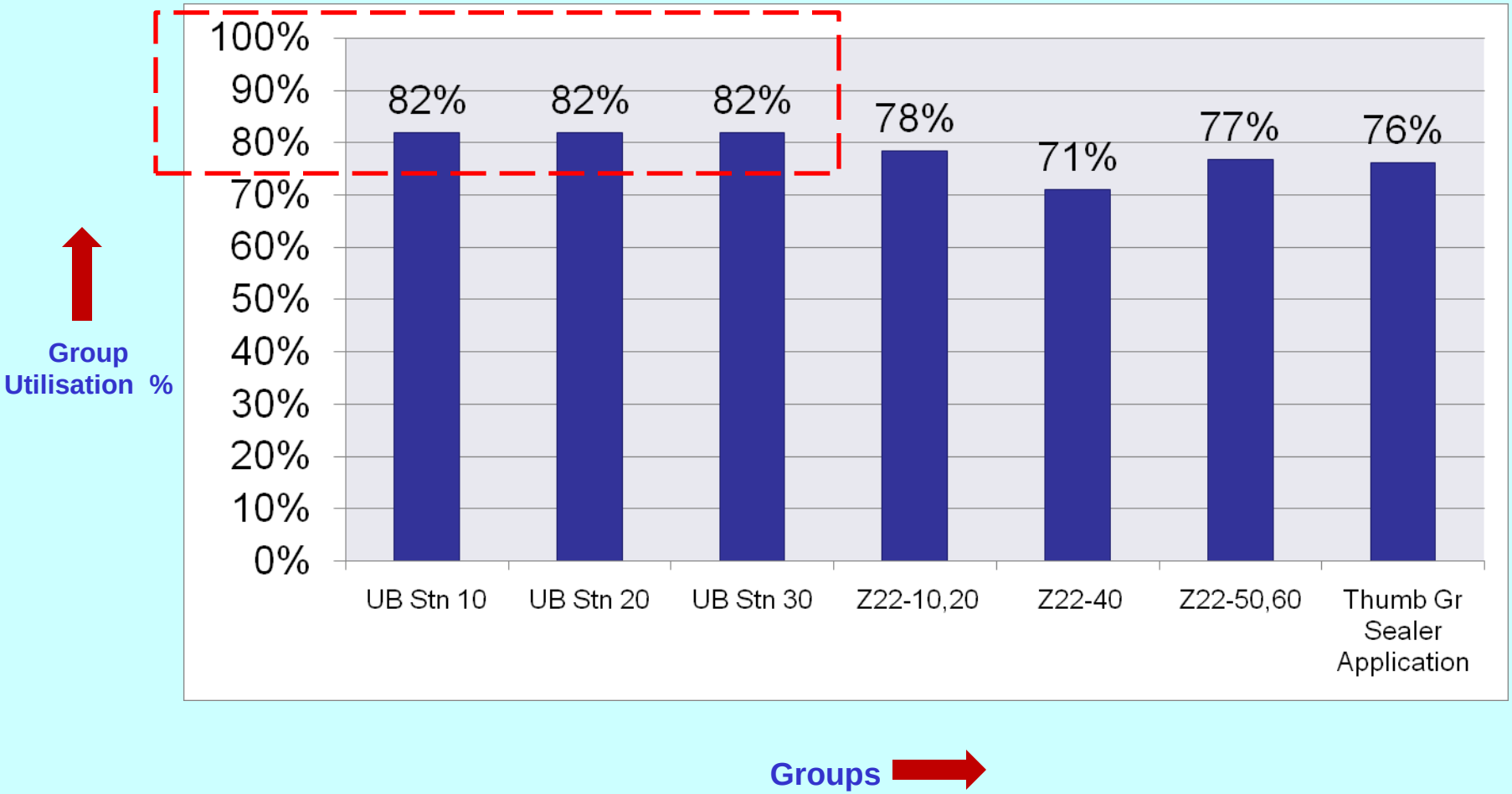
After Improvement :

Transformer & guns mounted on monorails which very less required effort

Step 7 Confirm Effect /Verify Results

7.1 Verify Results:

Framing line Under Body - Existing group Utilization



Step 7 Confirm Effect /Verify Results

7.1 Verify Results:

Framing Line Utilization After 25% Rise

Step 7 Confirm Effect /Verify Results

7.3Tangible & Intangible benefits:

Tangible Benefits

1. 25 % Productivity rise (92 to 114 Set up)
2. Work Content / NVA Reduction of 0.5min (Under Body #30)
3. Optimizes operator Utilization –(96 % to 82 %)
4. Annualise Manpower Cost Saving- 3 lacs

Intangible Benefits

1. Meeting requirement of forecast demands

Step 7 Confirm Effect /Verify Results

7.2 Full Scale Implementation plan:

Implementation Plan

SNO	Group	NVA /Work Content reduction in Min			Target	Responsibility	Status	Remarks
		WCR Required	WCR Identified	Effective WCR Implemented				
9	Framing Line – UB#30	1.2	1.2	1.2	31.12.12	Tushar	In process	After WCR : Current utilization :96 % Future Utilization :91.6 %

Step 8. Standardization

8.1 Updation of documents:

1. Most Element sheets

2. Work content Sheets

System is generating report: Please wait.....

MOST Element Report

Operation Group ID: 01 SBS FRM_F08

Operation No. : 53.024 SEALER APPLICATION (ASSEMBLY UNDER BODY & FRONT END) RH & LH OPERATOR REV2

Sr. No.		Element Description	Freq.	Div.	Men	Offline
1		GRASP SEALER GUN	1	1	2	1
	1	REACH, GRASP PART	1	1	1	No
2		WITHINREACH	1	1	2	1
	1	REACH	1	1	1	No
3		PUT SELAER GUN ON 3 LOCATION	3	1	2	1
	1	PUT THE PART	1	1	1	No
4		PRESS THE BUTTON OF SEALER GUN 3 TIMES	3	1	2	1
	1	PUSH PULL SWITCH / ROTATE KNOB, SLIDE / PUSH BUTTON / PUSH/PULL/LEVER/PIVOT <=12"	1	1	1	No
5		PT 2.5 SEC AT 3 LOCATION(1 LOCATIONS DELETED ON RH AND 1 ON LH DUE TO ECN)	7.5	1	2	1
	1	AUTOCYCLE TIME 1 SEC	1	1	1	No
6		WITHIN REACH PUT GUN ON STAND	1	1	2	1
	1	REACH, PUT PART	1	1	1	No

8.2 Training:

1. Modified element sheets are shared with concern operators

Step 8. Standardization

8.3 Additional Audit / Check

1. Work content sign off made compulsory for Project / EN

PROJECT/EN DELIVARABLES

CHANGE DESCRIPTION	
REASON	
OWNER	
CDMM	
EXP. IMPL. DATE	

R N	BUILD DELIVERABLES	DETAILS	C/O	lical	GRY	Remarks
1	Copmpatibility with 220 UPS (127 secs TACT)					
1	STATIC BUILD					
2	BOM VERIFICATION					
3	WORK CONTENT ESTIMATION					
4	IDENTIFICATION OF FTG					
5	OPERATOR TRAINING					
6	CONCERNS IDENTIFICATION					
7	COMMUNICATION OF CONCERNS TO THE DESIGNER					
8	PROCESS FLOW					
9	PROCESS SHEET					
10	PROCESS FMEA					
11	CONTROL PLAN					
12	SOS VALIDATION					
13	SOS UPDATION					
14	TPPS STUDY					
15	ANOVA-C AND PRODUCT VALIDATION					
16	ES TEST					
17	REPAIR STANDARD					
18	TRAVELLER CARD					
19	BUILD SHEET					
20	BUY OFF CHECK SHEET					
21	MSDS					
22	HIRA					

9. Summarize & Plan future actions

9.1 Conclusion

With the help of Effective Review mechanism, Project targets are met on time with fixed goal of 25% Productivity rise and Team morale improved

9. Summarize & Plan future actions

9.2 Horizontal Deployment:

Work content ReductionAction Plan & Status

SNO	Group	NVA /Work Content reduction in Min			Target	Responsibility	Status	Remarks
		WCR Required	WCR Identified	Effective WCR Implemented				
9	Framing Line – SCDC Respot	1.3	1.3	1.3	15.1.12	Rajesh	In process	Rocker Panel Automation

9. Summarize & Plan future actions

9.3 Lessons Learnt:

TGW (Things Gone Wrong) :

1. Too much analysis of video shooting.

TGR (Things Gone Right) :

1. Cross functional team formation and Approach
2. Effective Review Mechanism Every Week
3. Seniors involvement in the project themes and trials
4. Scope Identification method thru Video Analysis

Thank You!

T NO : 209301

Name : Rajesh Dhanokar

Dept : Scorpio Body Shop